

Lab 2 Activity

In this lab, we will create some visualizations of the `diamonds` dataset. Use Quarto to create a Word or pdf document of your results. When you are done (or when lab time runs out), check in with the TA to receive participation credit for the day. **This assignment is not graded aside from participation credit**

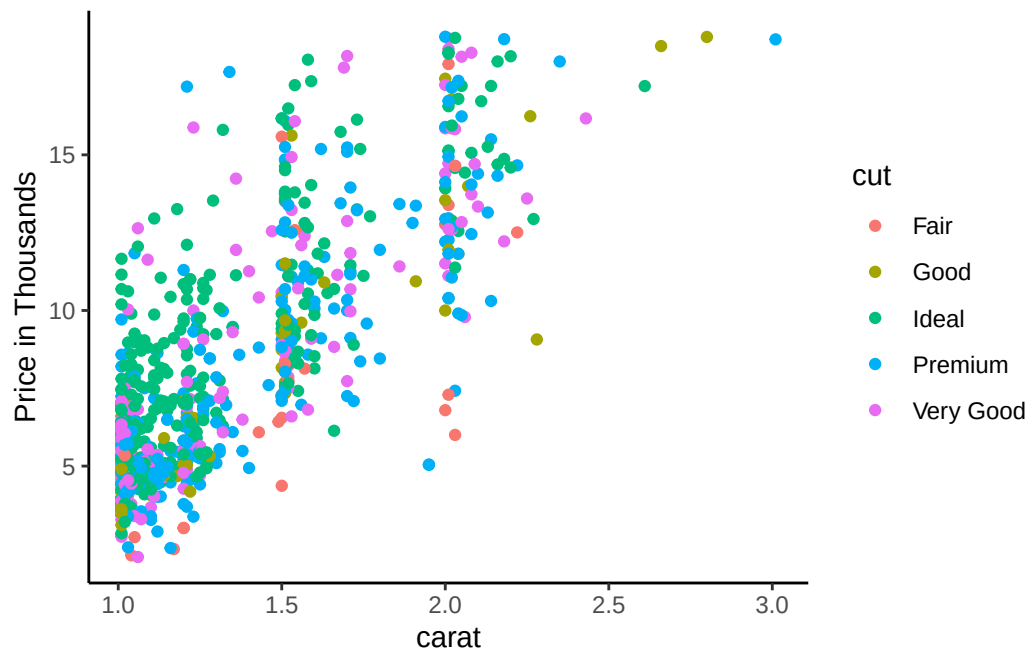
First make sure that you download the “diamonds.csv” dataset. Make sure that this file is in your working directory (you may want to change your working directory). See Lab 1 for help with this step.

```
library(tidyverse)
diamonds <- read.csv("diamonds.csv")
```

You can run the `?diamonds` command in your console to get a description of the variables in the dataset. Additionally, check out [this page](#) for even more diamond knowledge!

Recreate the following graphs to the best of your ability (feel free to start with any one you please). Use the `ggplot` help pages and the internet to help you! We don’t expect you reproduce every aspect of every plot today, but we want to give you some real-world problem-solving practice. After lab today, we’ll give you the source code for you to review on your own time.

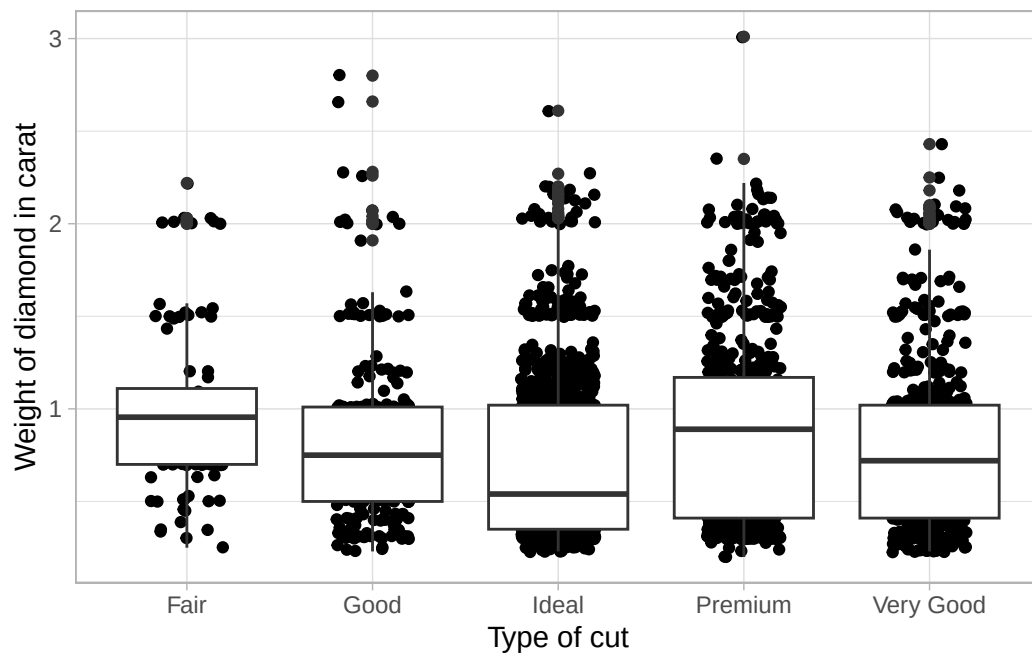
Plot 1



Plot 1 checklist:

- Filter only diamonds that are above 1 carat
- Transform `price` (in USD) to thousands of USD
- Plot different cuts in different colors
- Create a scatterplot with `price` on x-axis and `carat` on y-axis
- Make sure y axis label matches the plot
- Choose a plotting theme that omits the background grid lines

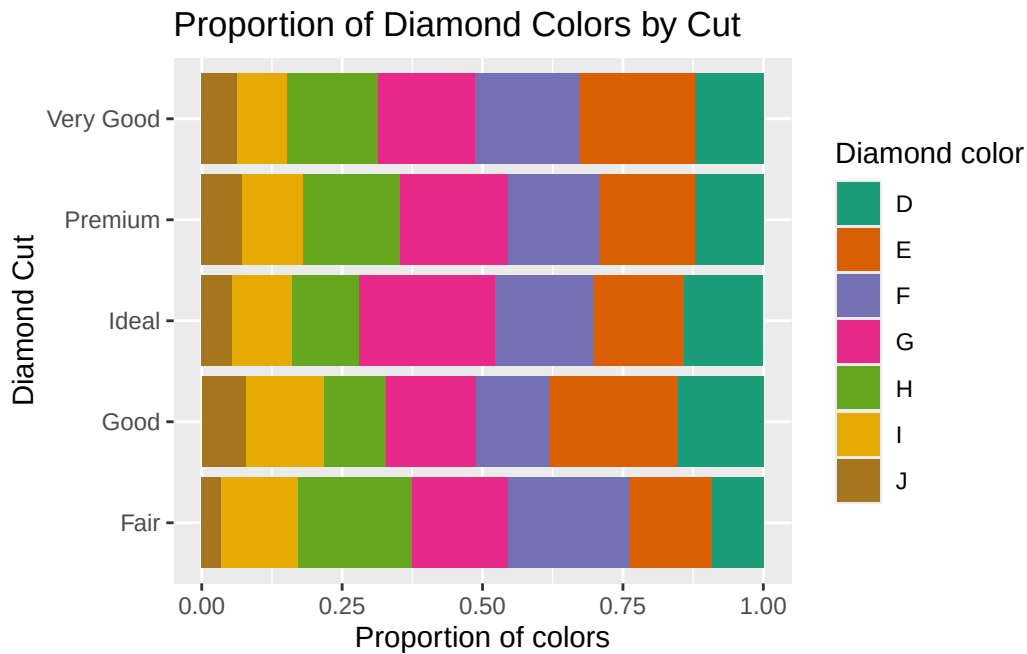
Plot 2



Plot 2 checklist

- cut should be converted to a factor type variable
- Create boxplots
- Create jittered scatterplots with width of .2
- Choose a theme that matches the picture
- Match x and y axis labels

Plot 3



Plot 3 checklist:

- For each `cut`, calculate the proportion of each colors. The `group_by()` and `summarise()` functions should be quite helpful to do this.
- Create a stacked bar plot that fills the plotting space
- Use different colors for different diamond colors
- Different cuts should be on different rows
- Match the axis and the title
- To match colors exactly, use the “Dark2” palette from the `RColorBrewer` package (see `scale_fill_brewer()` function in `ggplot2`)